Chromosome 17q12 Deletion Syndrome (HNF1B Microdeletion / Renal Cysts and Diabetes Syndrome): A Comprehensive Disease Characterization

MONDO: MONDO:0013797 · **OMIM:** 614527 (17q12 recurrent deletion syndrome) / 137920 (RCAD, *HNF1B*) · **Orphanet:** ORPHA:261265 (17q12 microdeletion) / ORPHA:93111 (RCAD) **Category:** Mendelian (autosomal-dominant contiguous-gene microdeletion / single-gene haploinsufficiency) **Prepared:** 2026-09-03 · Iterations 1–5 (final) · 8 confirmed findings, 35 references reviewed

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Summary

Chromosome 17q12 deletion syndrome is a recurrent, autosomal-dominant contiguous-gene microdeletion spanning approximately **1.4–1.8 Mb** at chromosome band 17q12. The deletion is generated by **non-allelic homologous recombination (NAHR)** between flanking low-copy repeats and removes one copy of **HNF1B** (hepatocyte nuclear factor 1 homeobox B) together with roughly **15 neighboring genes** (including *ACACA*, *ZNHIT3*, *CCL3L1*, *PIGW*, *AATF*, *DDX52*). Haploinsufficiency of the transcription factor *HNF1B* is the principal driver of the renal, pancreatic, hepatic, genital, and electrolyte phenotypes, while the co-deleted genes contribute the neurodevelopmental and neuropsychiatric burden (developmental delay, autism spectrum disorder, schizophrenia). The disorder is clinically equivalent, at the renal/endocrine level, to intragenic *HNF1B* mutations — together labeled **HNF1B-associated disease** or **Renal Cysts and Diabetes syndrome (RCAD)**, and the diabetes form is **MODY5**.

HNF1B is a master regulator of nephron, pancreas, biliary and Müllerian-duct development. It directly transactivates cystic-kidney genes **PKHD1** and **UMOD**, the distal-tubule magnesium gene **FXYD2**, and cooperates with developmental patterning genes (**pax2, wt1, pdx1, shh**) demonstrated in a zebrafish *vhnf1/hnf1b* model. Loss of one functional copy therefore produces a stereotyped multi-organ picture: **renal cysts/dysplasia with progressive tubulointerstitial**

chronic kidney disease (CKD stage 3–4 in ~44%, end-stage renal disease in ~21% of adults), **MODY5 diabetes with pancreatic hypoplasia** (usually insulin-requiring, ~79% on insulin at follow-up), **renal magnesium wasting/hypomagnesemia** (50–60%), **genital-tract Müllerian malformations**, **hyperuricemia/gout**, and **liver-enzyme abnormalities**. There is no clear genotype–phenotype correlation, consistent with haploinsufficiency, and expressivity is highly variable with incomplete penetrance.

Population prevalence of the 17q12 deletion is approximately **1 in 4,000 newborns**, with roughly **one-third arising de novo**. Diagnosis relies on chromosomal microarray/CNV sequencing (for the deletion) or *HNF1B* sequencing (for intragenic variants), guided by the validated **HNF1B clinical score** (AUC 0.78; a score <8 rules out disease with negative predictive value >99%). Management is entirely organ-directed and supportive — insulin for diabetes, magnesium repletion, urate-lowering therapy, nephrology/CKD care, and genetic counseling with prenatal and preimplantation testing options. No disease-specific or curative therapy exists.

Key Findings

Finding 1 — The 17q12 deletion is a recurrent NAHR-mediated CNV encompassing *HNF1B* and ~15 genes

The core lesion is a **recurrent ~1.4–1.8 Mb deletion** at 17q12, mediated by non-allelic homologous recombination between segmental duplications flanking the interval. In a large neurodevelopmental cohort, the deletion was detected in **18/15,749 patients versus 0/4,519 controls**, with follow-up enrichment in autism spectrum disorder (2/1,182) and schizophrenia (4/6,340) versus 0/47,929 controls (corrected $p = 7.37\times10^{-5}$) (PMID: 21055719). The deleted interval harbors *HNF1B* — "the gene responsible for renal cysts and diabetes syndrome (RCAD)" — plus approximately 15 genes, and the authors proposed that "one or more of the 15 genes in the deleted interval is dosage sensitive and essential for normal brain development and function," establishing the contiguous-gene syndrome model.

Reported deletion sizes cluster between **1.4 and 1.9 Mb** across cohorts (e.g., 1.494–1.66 Mb; 1.46 Mb; a prototypical 1.4 Mb interval `arr 17q12(34,850,785_36,248,926)x1`). The genes recurrently cited within the interval include *HNF1B, ACACA, ZNHIT3, CCL3L1, PIGW* (5 OMIM genes) and up to 17 protein-coding genes including *AATF* and *DDX52*. This dual architecture —

a single dosage-critical developmental transcription factor (*HNF1B*) plus additional dosage-sensitive neurodevelopmental genes — explains why the deletion phenotype is broader than that of intragenic *HNF1B* variants.

Finding 2 — Microdeletions, but not reciprocal microduplications, drive the diabetes phenotype

Dosage directionality matters. In UK Biobank (**450,993 individuals**), 11 microdeletions and 106 microduplications at 17q12 were identified; **microdeletions were strongly associated with diabetes**, whereas the reciprocal **microduplications were associated with renal disease but not diabetes** (PMID: 36109160). The paper concludes: "We demonstrate 17q12 microdeletions but not microduplications are associated with diabetes in a population-based cohort." This confirms that the diabetes/MODY5 component is specifically a **loss-of-dosage (haploinsufficiency)** phenomenon of *HNF1B*, and it distinguishes the clinical consequences of the reciprocal CNVs at this locus.

Finding 3 — HNF1B → FXYD2 axis explains renal magnesium wasting

HNF1B directly regulates **FXYD2**, the γ-subunit of the Na^+,K^+-ATPase, in the **distal convoluted tubule (DCT)**. HNF1B, together with cofactor **PCBD1**, co-stimulates the *FXYD2* promoter, and FXYD2 activity "is instrumental in Mg^{2+} reabsorption in the DCT" (PMID: 24204001). Consequently, **50–60% of ADTKD-HNF1B patients develop hypomagnesemia** (PMID: 26340261), and the biochemical signature is distinctive: "All patients presented with hypomagnesemia with a high fractional excretion of Mg^{2+} and hypocalciuria." Hypomagnesemia may be the **first clinical manifestation** of HNF1B disease, giving it a Gitelman-like electrolyte profile (hypomagnesemia, hypokalemic alkalosis, hypocalciuria). This provides a mechanistically-defined biomarker for the syndrome.

Finding 4 — Population prevalence ~1:4,000 newborns, ~one-third de novo

In a large newborn trio study (**12,252 MoBa trios**), the 17q12 deletion prevalence was estimated at **~1:4,000**, in the context of an overall recurrent neurodevelopmental CNV prevalence of ~0.48% (1 in 200) (PMID: 32778765). Approximately **34% (20/59) of recurrent NDD CNVs were de novo**. Prenatally, the detection rate among fetuses with urinary tract anomalies was ~6.5% (3/46), and other prenatal series report ~0.36% among fetuses undergoing CNV testing for ultrasound anomalies — of whom the vast majority present with

bilateral hyperechogenic kidneys. These figures anchor recurrence-risk counseling: an affected parent transmits with 50% probability, but a substantial fraction of index cases are new mutations.

Finding 5 — Quantitative phenotype frequencies and long-term renal prognosis

The largest adult cohort (**201 adults** with *HNF1B* molecular defects; Dubois-Laforgue 2017) quantifies the mature phenotype (PMID: 28420700):

Phenotype	Frequency	HPO term
Diabetes mellitus	159/201	HP:0000819
Renal cysts	122/166 (73%)	HP:0000107
CKD stage 3–4	75/169 (44%)	HP:0012622
End-stage renal disease	36/169 (21%)	HP:0003774
Diabetic retinopathy/neuropathy	46/114	HP:0000488 / HP:0000762
On insulin at follow-up	111/140 (79%)	—

"Chronic kidney disease stages 3-4 (CKD3-4) in 75 of 169 (44%), and end-stage renal disease (ESRD) in 36 of 169 (21%)" and "111 of 140 patients (79%) were treated with insulin at follow-up" quantify the two dominant clinical burdens. Molecularly, whole-gene deletion (i.e., the 17q12 deletion) and intragenic *HNF1B* mutations "each account for ~50% of all cases of HNF1B-associated disease," and importantly "there is no clear genotype-phenotype correlation, consistent with haploinsufficiency as the disease mechanism" (PMID: 25536396). This lack of genotype–phenotype correlation is a defining feature — the deletion and point mutations produce clinically indistinguishable renal/endocrine disease.

Finding 6 — The HNF1B score rationalizes genetic testing

Because the phenotype is protean, Faguer et al. developed a **17-item HNF1B score** (antenatal discovery, family history, and kidney/pancreas/liver/genital involvement). In a **433-individual cohort with 56 HNF1B cases**, "the HNF1B score efficiently and significantly discriminated between mutated and nonmutated cases (AUC 0.78)," and "the optimal cutoff threshold for the negative predictive value to rule out HNF1B mutations in a suspected individual was 8 (sensitivity 98.2%, specificity 41.1%, and negative predictive value over 99%)" (PMID:

24897035). The same work confirms "an autosomal-dominant inheritance, a 50% rate of de novo mutations, and a highly variable phenotype." A score below 8 effectively excludes disease, sparing unnecessary sequencing.

Finding 7 — HNF1B regulates PKHD1 and UMOD; biallelic loss drives chromophobe RCC

HNF1B directly regulates the cystic-kidney genes **PKHD1** (the ARPKD gene) and **UMOD** (uromodulin). In renal tumorigenesis, "biallelic HNF1beta inactivation was found in two of 12 chromophobe renal carcinomas by association of a germline mutation and a somatic gene deletion. In these cases, the expression of PKHD1 … and UMOD …, two genes regulated by HNF1beta, was turned off" (PMID: 15649945). This defines a co-regulated **HNF1B–PKHD1–UMOD** transcriptional cluster central to tubular/cystic biology. A broader survey of **130 kidney tumors** found decreased *HNF1B* expression associated with higher grade/stage in clear cell RCC, supporting that "in ccRCC and chRCC it may act in a tumour suppressive fashion" (PMID: 33051485). Clinically, this links the germline haploinsufficiency of the syndrome to a theoretical (though not established as high-frequency) renal-tumor predisposition via second-hit somatic inactivation.

Finding 8 — Zebrafish hnf1b (vhnf1) model recapitulates the syndrome via patterning-gene regulation

The developmental mechanism is directly demonstrated in a model organism. Insertional zebrafish **vhnf1 (hnf1b) mutants** show "formation of kidney cysts, underdevelopment of the pancreas and the liver, and reduction in size of the otic vesicles" — a striking recapitulation of human MODY5/RCAD (PMID: 11731484). Mechanistically, "vhnf1 is required for the proper expression of pdx1 and shh (sonic hedgehog) in the gut endoderm, pax2 and wt1 in the pronephric primordial, and valentino (val) in the hindbrain." This places *HNF1B* upstream of the master patterning genes for pancreas (**pdx1**), pronephros/kidney (**pax2, wt1**), and hindbrain (**valentino/mafba**), providing the causal bridge from transcription-factor haploinsufficiency to organ malformation.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **NAHR between flanking low-copy repeats at 17q12 → generates a recurrent ~1.4–1.8 Mb heterozygous deletion** (de novo in ~1/3, inherited AD in ~2/3).
2. The deletion **removes one copy of *HNF1B* plus ~15 contiguous genes → HNF1B haploinsufficiency** (≈50% dosage of the transcription factor).
3. Reduced HNF1B dosage → **failure to maintain normal transcription of direct target genes** in a tissue-specific manner. This branches:
4. **Kidney branch:** ↓ regulation of *PKHD1*, *UMOD*, and patterning genes *pax2/wt1* → **abnormal nephron/tubule development, tubular dilatation and cyst formation, dysplasia → progressive tubulointerstitial fibrosis → CKD → ESRD** (*inferred causal chain from direct-target regulation + zebrafish model + human frequencies*).
5. **DCT electrolyte branch:** ↓ HNF1B/PCBD1 co-stimulation of *FXYD2* → **impaired Na,K-ATPase γ-subunit function in the DCT → reduced Mg^{2+} reabsorption → renal Mg wasting → hypomagnesemia with high fractional Mg excretion and hypocalciuria** (*demonstrated*).
6. **Pancreas branch:** ↓ *pdx1/shh*-dependent pancreatic patterning → **pancreatic (body/tail) hypoplasia → β-cell deficiency + exocrine insufficiency → MODY5 diabetes, usually insulin-requiring** (*inferred from model + clinical correlation*).
7. **Hepatobiliary branch:** HNF1B loss in biliary/hepatic epithelium → **liver-enzyme elevations, biliary abnormalities** (*observed clinically*).
8. **Genital branch:** ↓ Müllerian-duct development → **uterine/genital-tract malformations (e.g., bicornuate/incomplete uterus), genital anomalies** (*observed clinically*).
9. **Metabolic branch:** tubular dysfunction → **hyperuricemia/gout** (*observed clinically*).
10. **Neurodevelopmental branch (co-deleted genes, not HNF1B):** haploinsufficiency of ≥1 of the other ~15 dosage-sensitive genes → **developmental delay, learning difficulty, autism spectrum disorder, schizophrenia risk** (*inferred; specific gene not resolved*).
11. In a subset of renal epithelial cells, a **somatic second hit inactivating the remaining *HNF1B* allele → biallelic HNF1B loss → loss of PKHD1/UMOD expression → contribution to chromophobe/clear-cell renal carcinoma** (*demonstrated in tumors; population-level cancer risk in syndrome not quantified*).

Upstream vs downstream

- **Most upstream:** the NAHR deletion and resulting *HNF1B* haploinsufficiency (and co-deletion of neurodevelopmental genes).
- **Intermediate:** dysregulation of direct targets (*PKHD1*, *UMOD*, *FXYD2*) and patterning genes (*pax2*, *wt1*, *pdx1*, *shh*).
- **Downstream:** organ malformation (cysts, pancreatic hypoplasia, Müllerian defects), tubular electrolyte handling defects, and progressive fibrosis/CKD.

Text schematic

```
   NAHR at 17q12 low-copy repeats
               │
   ~1.4–1.8 Mb heterozygous deletion
         ┌─────┴──────────────────┐
   HNF1B haploinsufficiency    ~15 co-deleted genes
         │                            │
   ┌─────┼───────┬────────┬──────┐    └─► neurodevelopment:
   ▼     ▼       ▼        ▼      ▼          DD / ASD / schizophrenia
PKHD1  UMOD   FXYD2   pax2/wt1  pdx1/shh
   │     │       │        │        │
 cysts/dysplasia   ↓Mg²⁺     kidney    pancreas
   → CKD → ESRD  reabsorption  malform.  hypoplasia
                     │                      │
             hypomagnesemia          MODY5 diabetes
             (Gitelman-like)        (insulin-requiring)
```

Ontology annotations

- **Gene:** HNF1B (HGNC:11630; NCBI Gene 6928; UniProt P35680; OMIM 189907)
- **GO biological process:** regionalization (GO:0003002), metanephros development (GO:0001656), pronephros development (GO:0048793), mesonephric tubule development (GO:0072164), endocrine pancreas development (GO:0031018), positive regulation of transcription by RNA polymerase II (GO:0045944), magnesium ion transmembrane transport (GO:1903830).
- **GO cellular component:** nucleus (GO:0005634); basolateral plasma membrane / Na,K-ATPase complex for FXYD2 (GO:0005890).
- **Cell types (CL):** kidney distal convoluted tubule epithelial cell (CL:1000849), kidney collecting duct epithelial cell (CL:1000454), pancreatic A/B cell (CL:0000171 / CL:0000169), hepatocyte (CL:0000182), epithelial cell (CL:0000066).

- **UBERON anatomy:** kidney (UBERON:0002113), renal tubule/nephron (UBERON:0001231 / UBERON:0001285), distal convoluted tubule (UBERON:0001292), pancreas (UBERON:0001264), liver (UBERON:0002107), uterus (UBERON:0000995), Müllerian duct (UBERON:0003890).
- **CHEBI:** magnesium(2+) (CHEBI:18420), uric acid/urate (CHEBI:27226 / CHEBI:17775), D-glucose (CHEBI:17234), insulin (peptide hormone).

Evidence Base

PMID	Title (abbrev.)	Evidence type	How it supports the model
21055719	Deletion 17q12 confers high risk of autism/ schizophrenia	Human case-control genomics	Defines recurrent ~1.4 Mb deletion, ~15-gene content, case enrichment (F001)
36109160	17q12 microduplications contribute to renal disease not diabetes	Human population cohort (UK Biobank)	Dosage directionality: deletion → diabetes, duplication → renal (F002)
24204001	PCBD1 mutations cause hypomagnesemia	In vitro / human genetics	Establishes HNF1B/ PCBD1→FXYD2 axis for Mg handling (F003)
26340261	Hypomagnesemia as first manifestation of ADTKD-HNF1B	Human case series	Quantifies hypomagnesemia (50–60%) and its biochemical signature (F003)
30175537	Renal Mg handling, FXYD2, Na,K-ATPase	Review / mechanism	Mechanistic detail on FXYD2 regulatory role (supports F003)
35894287	Genetic spectrum of Gitelman-like syndromes	Review	Places HNF1B among Gitelman-like DCT electrolyte disorders (supports F003)
32778765	Population prevalence of recurrent CNVs in newborns	Human newborn trios	Prevalence ~1:4,000; ~34% de novo (F004)
28420700	201 adults with HNF1B — long-term prognosis	Human clinical cohort	Quantitative phenotype frequencies, renal/diabetes prognosis (F005)
25536396	HNF1B disease — expanding spectrum	Review	~50/50 deletion vs intragenic; haploinsufficiency, no genotype-phenotype correlation (F005)
24897035	HNF1B score for patient selection	Human diagnostic study	Score AUC 0.78; cutoff 8, NPV >99% (F006)
15649945	Germline HNF1α/β mutations in RCC	Human tumor genetics	Biallelic HNF1B loss in chromophobe RCC; PKHD1/UMOD regulation (F007)

PMID	Title (abbrev.)	Evidence type	How it supports the model
33051485	HNF1B in 130 kidney tumors	Human tumor genomics	Tumor-suppressive role in ccRCC/chRCC (F007)
11731484	Zebrafish vhnf1 regulates gut/pronephros/hindbrain	Model organism	Phenotype recapitulation + downstream patterning targets (F008)
20378824	Reduced Notch → renal cysts, microadenomas	Mouse model	Contextual: cyst/tumor biology intersecting HNF1B (TCF2) pathway
30525249	Acetylation drives HNF1β stability	In vitro	Protein-level regulation of HNF1B stability (mechanistic detail)

Multiple recent case reports (PMID: 41924323, 38432894, 38044981, 41694676, 37799485) corroborate the multi-organ spectrum including hyperuricemia, hypomagnesemia, muscle-mass loss, nephrocalcinosis, and elevated liver enzymes (MODY5). Prenatal series (PMID: 35232906, 38957807, 37212013, 32219821, 41999034) consistently identify **bilateral hyperechogenic/cystic kidneys** as the dominant prenatal presentation and confirm chromosomal microarray/CNV-seq as the pivotal diagnostic tool. A novel HNF1B-disrupting **inversion** (PMID: 41703530) and an intragenic hotspot deletion p.(Gly239del) (PMID: 31498910) broaden the variant spectrum.

Section-by-Section Synthesis

1. Disease Information

17q12 deletion syndrome is a recurrent contiguous-gene microdeletion causing a multisystem disorder dominated by renal, endocrine (diabetes), and neurodevelopmental features. **Identifiers:** MONDO:0013797; OMIM 614527 (deletion) and 137920 (RCAD); Orphanet ORPHA:261265 / ORPHA:93111; MeSH aligns with "Chromosome Deletion" + "HNF1B." **Synonyms:** HNF1B microdeletion syndrome; Renal Cysts and Diabetes syndrome (RCAD); MODY5; HNF1B-associated disease; ADTKD-HNF1B (when tubulointerstitial). Information derives from **aggregated disease-level resources** (OMIM/Orphanet) and **individual clinical cohorts/case reports**, not EHR-scale phenotyping.

2. Etiology

Primary cause: heterozygous 17q12 deletion (NAHR-mediated) or, in ~50% of HNF1B-disease cases, intragenic *HNF1B* variants. **Genetic risk:** essentially the deletion/variant itself; there are no well-established susceptibility modifier loci. **Environmental risk factors, protective factors, and gene-environment interactions are not established** for this Mendelian disorder — onset and organ involvement track the germline lesion, not exposures. De novo occurrence (~1/3 of deletions; ~50% of intragenic variants) means absence of family history does not exclude the diagnosis.

3. Phenotypes

Key phenotypes with HPO suggestions and frequencies (from F003, F005): renal cysts (HP:0000107, 73%), CKD (HP:0012622, ~44% stage 3–4), ESRD (HP:0003774, ~21%), diabetes mellitus/MODY (HP:0000819 / HP:0004904, majority), hypomagnesemia (HP:0002917, 50–60%), pancreatic hypoplasia (HP:0002983), Müllerian/uterine malformation (HP:0000130 / HP:0000132), hyperuricemia/gout (HP:0002149 / HP:0001997), elevated liver enzymes (HP:0002910), developmental delay (HP:0001263), autism (HP:0000717), and hypocalciuria (HP:0003169). **Onset spans prenatal (hyperechogenic kidneys) through childhood/adult (MODY5 typically diagnosed in adolescence/early adulthood).** Severity and progression are **highly variable** with incomplete penetrance; renal disease is typically **progressive**. Quality-of-life impact is driven chiefly by CKD/dialysis burden, insulin-dependent diabetes, and (in deletion cases) neurodevelopmental/psychiatric morbidity; formal EQ-5D/SF-36 data specific to the syndrome were not identified.

4. Genetic/Molecular Information

Causal gene: *HNF1B* (HGNC:11630; OMIM 189907). **Variant classes:** whole-gene deletion (the 17q12 CNV, ~50%), and intragenic pathogenic/likely-pathogenic variants (missense, nonsense, frameshift, splice, in-frame deletions such as p.(Gly239del) in the DNA-binding domain hotspot; also point substitutions e.g. C295R) per ACMG/AMP. A novel **inversion disrupting HNF1B** (GRCh38:17:g.36934029_37729559inv) has also been reported. **Functional consequence:** loss of function → haploinsufficiency; **no clear genotype–phenotype correlation. Origin:** germline (frequently de novo). **Chromosomal abnormality:** recurrent 17q12 interstitial deletion, ~1.4–1.8 Mb. **Modifier genes/epigenetics:** not established, though HNF1β protein stability is modulated post-translationally by acetylation (PMID: 30525249).

5. Environmental Information

Not applicable as a cause. No toxin, infectious agent, or lifestyle factor is established in disease causation. Standard diabetes/CKD lifestyle management applies to complication control but does not modify the germline etiology.

6. Mechanism / Pathophysiology

Presented as the ordered causal chain above. Core: *HNF1B* haploinsufficiency → dysregulation of direct targets **PKHD1, UMOD, FXYD2** and patterning genes **pax2, wt1, pdx1, shh** → renal cysts/dysplasia + tubulointerstitial fibrosis, DCT magnesium wasting, pancreatic hypoplasia, and Müllerian defects; co-deleted genes add neurodevelopmental risk. Molecular/omics-specific profiling of the syndrome (transcriptomics, proteomics, single-cell) is limited; the strongest mechanistic evidence is transcription-factor–target regulation plus the zebrafish model.

7. Anatomical Structures Affected

Primary organs: kidney (UBERON:0002113; bilateral, often symmetric), pancreas (UBERON:0001264), liver/biliary tract (UBERON:0002107), female genital tract/uterus (UBERON:0000995). **Body systems:** urinary, endocrine, hepatobiliary, reproductive, and (via co-deleted genes) central nervous system. **Tissue/cell level:** renal tubular epithelium — especially **DCT epithelial cells** (CL:1000849) and collecting duct — pancreatic islet and acinar cells, hepatocytes/cholangiocytes, Müllerian-duct epithelium. **Subcellular:** nucleus (transcription factor, GO:0005634); DCT basolateral Na,K-ATPase complex (FXYD2, GO:0005890). **Lateralization:** renal involvement is typically **bilateral**.

8. Temporal Development

Onset: congenital/prenatal renal structural abnormality (hyperechogenic kidneys detectable on second-trimester ultrasound) through adolescent/adult-onset diabetes. **Course:** renal disease is chronic and **progressive** toward CKD/ESRD; diabetes is progressive and usually insulin-requiring. **Critical periods:** fetal organogenesis (kidney/pancreas/Müllerian development) is the vulnerable window; postnatally, management targets complication prevention. Disease is **lifelong**; no spontaneous remission.

9. Inheritance and Population

Inheritance: autosomal dominant; **~50% de novo** for intragenic variants and **~1/3 de novo** for deletions. **Penetrance:** incomplete and variable; **expressivity highly variable** even within families (multigenerational reports show renal cysts, stones, diabetes, pancreatic dysfunction in different relatives). **Prevalence:** ~1:4,000 newborns for the deletion; HNF1B disease overall is a leading monogenic cause of developmental kidney disease. **Sex:** both sexes affected; females may present additionally with Müllerian anomalies. Founder effects, consanguinity, and anticipation are **not features** (dominant, often de novo).

10. Diagnostics

Genetic testing is definitive: chromosomal microarray (CMA)/CNV-seq detects the deletion; *HNF1B* sequencing (single-gene or panel) detects intragenic variants; whole-genome sequencing can resolve complex structural variants (e.g., inversion). **Clinical labs/ biomarkers:** hypomagnesemia with high fractional Mg excretion + hypocalciuria (distinctive); hyperuricemia; elevated liver enzymes; abnormal glucose/HbA1c; anti-GAD negativity helps distinguish MODY5 from type 1 diabetes. **Imaging:** renal ultrasound/MRI showing bilateral cysts, hyperechogenic/dysplastic kidneys; pancreatic imaging showing body/tail hypoplasia. **Clinical criterion/tool:** the **HNF1B score** (cutoff 8; NPV >99%) selects candidates for testing. **Differential diagnosis:** ADPKD, ARPKD (differentiated by targeted sequencing and family pattern), other ADTKD subtypes, Gitelman syndrome (for the electrolyte picture), and type 1 diabetes (for MODY5). **Screening:** prenatal CMA for fetuses with bilateral echogenic kidneys; cascade family testing.

11. Outcome/Prognosis

Renal: ~44% reach CKD 3–4 and ~21% ESRD in adulthood — the principal driver of morbidity and the main determinant of life expectancy (dialysis/transplant needs). **Diabetes:** usually insulin-requiring (~79%), with risk of microvascular complications. **Overall mortality** is not sharply elevated with modern renal replacement/diabetes care, but the disorder is chronic and lifelong. **Prognostic factors:** degree of renal impairment at diagnosis, rate of eGFR decline, and diabetes control. There is a theoretical renal-tumor consideration (chromophobe/clear-cell RCC via biallelic HNF1B loss), though population-level cancer risk in the syndrome is not established.

12. Treatment

No disease-specific or curative therapy. Management is organ-directed and supportive: - **Diabetes/MODY5:** insulin is the mainstay (NCIT: Insulin Therapy); a subset respond to sulfonylureas/repaglinide (~29/51 tested in the 201-adult cohort), but most are insulin-dependent at follow-up. - **Electrolytes:** oral/IV **magnesium** repletion for hypomagnesemia (CHEBI:18420); potassium citrate for stones/tubular acidosis. - **Hyperuricemia/gout:** urate-lowering therapy (e.g., allopurinol; NCIT: Allopurinol). - **CKD:** standard nephroprotection, and renal replacement (dialysis/transplant) for ESRD (NCIT: Renal Dialysis, Kidney Transplantation). - **Structural/genital anomalies:** surgical correction as indicated. - **Neurodevelopmental:** educational and behavioral support. No gene therapy, RNA therapy, or targeted molecular therapy exists; pharmacogenomics is limited to the sulfonylurea-responsiveness observation.

13. Prevention

No primary prevention (germline etiology). **Secondary prevention:** prenatal detection (CMA for echogenic kidneys), cascade family testing, and early metabolic/renal surveillance. **Genetic counseling** is central — 50% transmission risk from an affected parent, high de novo rate, variable expressivity; **preimplantation genetic testing (PGT)** and prenatal diagnosis are available for at-risk families. **Tertiary prevention:** aggressive CKD and diabetes complication management.

14. Other Species / Natural Disease

Orthologs: zebrafish *hnf1b/vhnf1*, mouse *Hnf1b* (NCBI Gene 21410). No prominent naturally-occurring companion-animal disease is catalogued; relevance is chiefly experimental. Evolutionary conservation of HNF1B's developmental role (kidney/pancreas/hindbrain patterning) is strong across vertebrates.

15. Model Organisms

- **Zebrafish** *vhnf1/hnf1b* insertional mutant — recapitulates kidney cysts, pancreatic/hepatic hypoplasia, reduced otic vesicles; reveals downstream targets *pdx1, shh, pax2, wt1, valentino* (PMID: 11731484). Strong developmental recapitulation; limitation — does not model adult-onset diabetes progression or human neuropsychiatric features.
- **Mouse** *Hnf1b* conditional/knockout models — used to study nephrogenesis and cystogenesis; homozygous null is embryonic lethal, requiring conditional/tissue-specific

approaches. A reduced-Notch mouse model produced renal cysts and papillary microadenomas with links to TCF2/HNF1β biology (PMID: 20378824).

- **Cellular/in vitro** — HNF1β-expressing lines (ES2, HEPG2, HK2) used to study protein stability/acetylation and target-gene regulation.

Limitations and Knowledge Gaps

1. **Neurodevelopmental gene not resolved.** The specific dosage-sensitive gene(s) among the ~15 co-deleted loci responsible for autism/schizophrenia/developmental delay remain unidentified; the contiguous-gene model is supported statistically but not gene-resolved.
2. **No genotype–phenotype correlation** limits prognostic precision — the deletion vs point mutation, and specific variants, do not reliably predict organ burden or severity.
3. **Penetrance/expressivity quantitatively uncertain.** Incomplete penetrance is documented, but robust penetrance estimates per organ system by variant type are lacking.
4. **Omics profiling of the human syndrome is sparse** — transcriptomic/proteomic/single-cell datasets specific to HNF1B-deleted human kidney/pancreas are limited; most mechanism is inferred from target-gene regulation and model organisms.
5. **Cancer risk unquantified.** Biallelic HNF1B loss occurs in chromophobe RCC, but the lifetime renal-tumor risk for germline 17q12-deletion carriers is not established.
6. **Quality-of-life and mortality data** specific to the syndrome (as opposed to CKD/diabetes generally) are limited; no syndrome-specific EQ-5D/SF-36 datasets were identified.
7. **Ascertainment bias.** Much of the cohort data comes from nephrology/genetics referral populations, likely over-representing severe renal phenotypes.

Proposed Follow-up Experiments / Actions

1. **Resolve the neurodevelopmental driver** via dosage analysis of individual 17q12 genes (e.g., CRISPR dosage models, or association testing of atypical smaller deletions) to pinpoint the ASD/schizophrenia gene(s).
2. **Prospective natural-history registry** stratified by variant type (deletion vs intragenic) to derive organ-specific penetrance, eGFR-decline trajectories, and mortality — addressing the genotype–phenotype and prognosis gaps.

3. **Single-cell/spatial transcriptomics** of HNF1B-deficient human kidney organoids and patient biopsies to map cell-type-specific dysregulation of PKHD1/UMOD/FXYD2 and validate the causal chain in human tissue.
4. **Quantify renal-tumor risk** through long-term surveillance imaging in a large carrier cohort and molecular characterization of any tumors for second-hit HNF1B inactivation.
5. **Pharmacogenomic trial of sulfonylurea/repaglinide responsiveness** to identify predictors of insulin-sparing response in HNF1B-MODY5.
6. **Systematic QoL assessment** (EQ-5D/PROMIS) across renal, endocrine, and neurodevelopmental domains to inform holistic management.
7. **Magnesium-repletion outcome study** to determine whether early correction of hypomagnesemia alters renal or metabolic trajectory.

Report compiled from 8 confirmed findings and 35 reviewed papers over a 5-iteration autonomous investigation. Evidence types span human clinical cohorts, human population genomics, tumor genetics, in vitro studies, and model-organism (zebrafish, mouse) work, as annotated in the Evidence Base.
